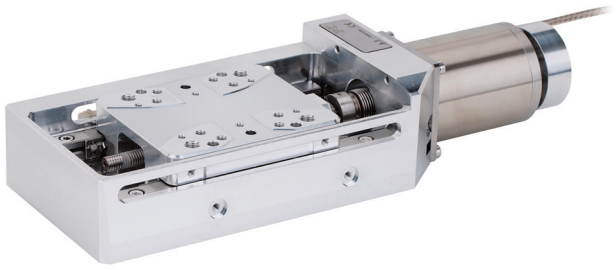
Vacuum-Compatible Precision Linear Stage

Compact Design, for Loads to 10 kg



L-509 V6 • V7 • V9

- Vacuum compatible up to 10^{-9} hPa
- Travel ranges from 26 to 102 mm (1" to 4")
- Repeatability to 0.1 µm
- DC gear motor, stepper motor
- Direction-sensing reference switch
- Multi-axis setups possible

Product overview
High travel accuracy and load capacity due to crossed roller guides with anti-creep system. Precision ball screw with 1 mm pitch. Compact design. Stress-relieved aluminum base for high stability. Limit switch. Noncontact optical reference switch with direction sensing in the middle of the travel range.

Reduced outgassing especially for use in vacuum environments
Most notably, for the lubrication of the guides and the spindle a material with particularly low outgassing is used. This allows a bakeout temperature of up to 80° C. With this and other modifications of the PI V6 vacuum class, this positioner can be used up to 10^{-6} hPa.

Higher quality for applications in high vacuum
In this case, special components suitable for high vacuum are used, such as motor, measuring system, and limit switches. In addition, the surface is made of uncoated aluminum and the spindles and guides are made of stainless steel. With this, the PI V7 vacuum class is reached, which means up to 10^{-7} hPa.

The highest quality for special requirements in ultrahigh vacuum
In this vacuum class, the product is optimized in all details thanks to components suitable for ultrahigh vacuum in order to reach PI V9 vacuum class up to 10^{-9} hPa. Among other things, only selected types of plastic are used for this purpose and bakeout temperatures of 120 °C with measuring system, respectively 150 °C without measuring system are applied.

Motor types available
- DC motor with gearhead for high torques and resolution at low motor power, to 10^{-6} hPa
- 2-phase stepper motor for low velocity and high resolution, to 10^{-9} hPa

Types of position measuring
- Without encoder (open loop)
- Integrated rotary encoder on the motor shaft

Application fields
Vacuum chambers. Research. Beamline instrumentation. Microscopy. Electronics assembly and inspection.

Motion	Unit	Tolerance	L-509.021100V6	L-509.031100V6	L-509.051100V6	L-509.024132V6	L-509.034132V6	L-509.054132V6	L-509.021300V7	L-509.031300V7
Active axes			X	X	X	X	X	X	X	X
Travel range in X	mm		26	52	102	26	52	102	26	52
Maximum velocity in X, unloaded	mm/s		10	10	10	1.5	1.5	1.5	10	10
Straightness error in Y (straightness)	µm	Typ.	±2	±4	±8	±2	±4	±8	±2	±4
Straightness error in Z (flatness)	µm	Typ.	±2	±4	±8	±2	±4	±8	±2	±4
Angular error around Y (pitch)	µrad	Typ.	±60	±90	±120	±60	±90	±120	±60	±90
Angular error around Z (yaw)	µrad	Typ.	±60	±90	±120	±60	±90	±120	±60	±90

Positioning	Unit	Tolerance	L-509.021100V6	L-509.031100V6	L-509.051100V6	L-509.024132V6	L-509.034132V6	L-509.054132V6	L-509.021300V7	L-509.031300V7
Minimum incremental motion in X	µm	Typ.	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
Unidirectional repeatability in X	µm	Typ.	±0.15	±0.15	±0.15	±0.05	±0.05	±0.05	±0.15	±0.15
Limit switches			Forked photoelectric sensor, N/C contact, 5 V, NPN	Forked photoelectric sensor, N/C contact, 5 V, NPN	Forked photoelectric sensor, N/C contact, 5 V, NPN	Forked photoelectric sensor, N/C contact, 5 V, NPN	Forked photoelectric sensor, N/C contact, 5 V, NPN	Forked photoelectric sensor, N/C contact, 5 V, NPN	Mechanical	Mechanical
Motor encoder			—	—	—	Incremental rotary encoder	Incremental rotary encoder	Incremental rotary encoder	—	—
Motor encoder: Sensor signal			—	—	—	A/B quadrature, RS-422	A/B quadrature, RS-422	A/B quadrature, RS-422	—	—

Drive properties	Unit	Tolerance	L-509.021100V6	L-509.031100V6	L-509.051100V6	L-509.024132V6	L-509.034132V6	L-509.054132V6	L-509.021300V7	L-509.031300V7
Drive type			2-phase stepper motor	2-phase stepper motor	2-phase stepper motor	DC gear motor	DC gear motor	DC gear motor	2-phase stepper motor	2-phase stepper motor
Nominal voltage	V		24	24	24	12	12	12	24	24
Maximum power consumption	W		10	10	10	—	—	—	10	10
Motor resolution	Full steps/rev.		200	200	200	—	—	—	200	200
Drive force in positive direction of motion in X	N	Typ.	60	60	60	60	60	60	60	60
Drive force in negative direction of motion in X	N	Typ.	60	60	60	60	60	60	60	60
Resistance phase-phase	Ω	Typ.	3.3	3.3	3.3	4.09	4.09	4.09	1.7	1.7
Inductance phase-phase	mH		2.8	2.8	2.8	0.18	0.18	0.18	3	3
Back EMF, phase-phase, rotational	V/kRPM	Max.	—	—	—	1.68	1.68	1.68	—	—

Mechanical properties	Unit	Tolerance	L-509.021100V6	L-509.031100V6	L-509.051100V6	L-509.024132V6	L-509.034132V6	L-509.054132V6	L-509.021300V7	L-509.031300V7
Permissible push force in Y	N	Max.	50	50	50	50	50	50	50	50
Permissible push force in Z	N	Max.	100	100	100	100	100	100	100	100
Permissible torque in θX	N·m	Max.	30	30	30	30	30	30	30	30
Permissible torque in θY	N·m	Max.	25	25	25	25	25	25	25	25
Permissible torque in θZ	N·m	Max.	20	20	20	20	20	20	20	20
Holding force in X, passive	N		50	50	50	—	—	—	50	50
Moved mass in X, unloaded	g		220	220	220	220	220	220	220	220
Drive screw type			Ball screw	Ball screw	Ball screw	Ball screw	Ball screw	Ball screw	Ball screw	Ball screw
Drive screw pitch	mm		1	1	1	1	1	1	1	1
Gear ratio i			—	—	—	2401 : 81	2401 : 81	2401 : 81	—	—
Guide			Crossed roller guide	Crossed roller guide	Crossed roller guide	Crossed roller guide	Crossed roller guide	Crossed roller guide	Crossed roller guide	Crossed roller guide
Overall mass	g		1400	1500	1700	1100	1200	1500	1000	1100
Material			Aluminum, steel	Aluminum, steel	Aluminum, steel	Aluminum, steel	Aluminum, steel	Aluminum, steel	Aluminum, steel	Aluminum, steel

Miscellaneous	Unit		L-509.021100V6	L-509.031100V6	L-509.051100V6	L-509.024132V6	L-509.034132V6	L-509.054132V6	L-509.021300V7	L-509.031300V7
Operating temperature range	°C		5 to 40	5 to 40	5 to 40	5 to 40	5 to 40	5 to 40	5 to 40	5 to 40
Vacuum class	hPa		10^{-6}	10^{-6}	10^{-6}	10^{-6}	10^{-6}	10^{-6}	10^{-7}	10^{-7}
Maximum bakeout temperature	°C		80	80	80	80	80	80	80	80
Connector			HD D-sub 26 (m)	HD D-sub 26 (m)	HD D-sub 26 (m)	HD D-sub 26 (m)	HD D-sub 26 (m)	HD D-sub 26 (m)	D-sub 15 (f)	D-sub 15 (f)
Cable length	m		2	2	2	2	2	2	2	2
Recommended controllers/ drivers			C-663.12 C-885 with C-663.12C885 G-901 G-910 ACS modular controller	C-663.12 C-885 with C-663.12C885 G-901 G-910 ACS modular controller	C-663.12 C-885 with C-663.12C885 G-901 G-910 ACS modular controller	C-863 C-885 with C-863.20C885 C-884 G-901 G-910 ACS modular controller	C-863 C-885 with C-863.20C885 C-884 G-901 G-910 ACS modular controller	C-863 C-885 with C-863.20C885 C-884 G-901 G-910 ACS modular controller	C-663.12 C-885 with C-663.12C885 G-901 G-910 ACS modular controller	C-663.12 C-885 with C-663.12C885 G-901 G-910 ACS modular controller

Motion	Unit	Tolerance	L-509.051300V7	L-509.031600V9	L-509.051600V9
Active axes			X	X	X
Travel range in X	mm		102	52	102
Maximum velocity in X, unloaded	mm/s		10	10	10
Straightness error in Y (straightness)	µm	Typ.	±8	±4	±8
Straightness error in Z (flatness)	µm	Typ.	±8	±4	±8
Angular error around Y (pitch)	µrad	Typ.	±120	±90	±120
Angular error around Z (yaw)	µrad	Typ.	±120	±90	±120

Positioning	Unit	Tolerance	L-509.051300V7	L-509.031600V9	L-509.051600V9
Minimum incremental motion in X	µm	Typ.	0.1	0.1	0.1
Unidirectional repeatability in X	µm	Typ.	±0.15	±0.15	±0.15
Limit switches			Mechanical	Mechanical	Mechanical
Motor encoder			—	—	—
Motor encoder: Sensor signal			—	—	—

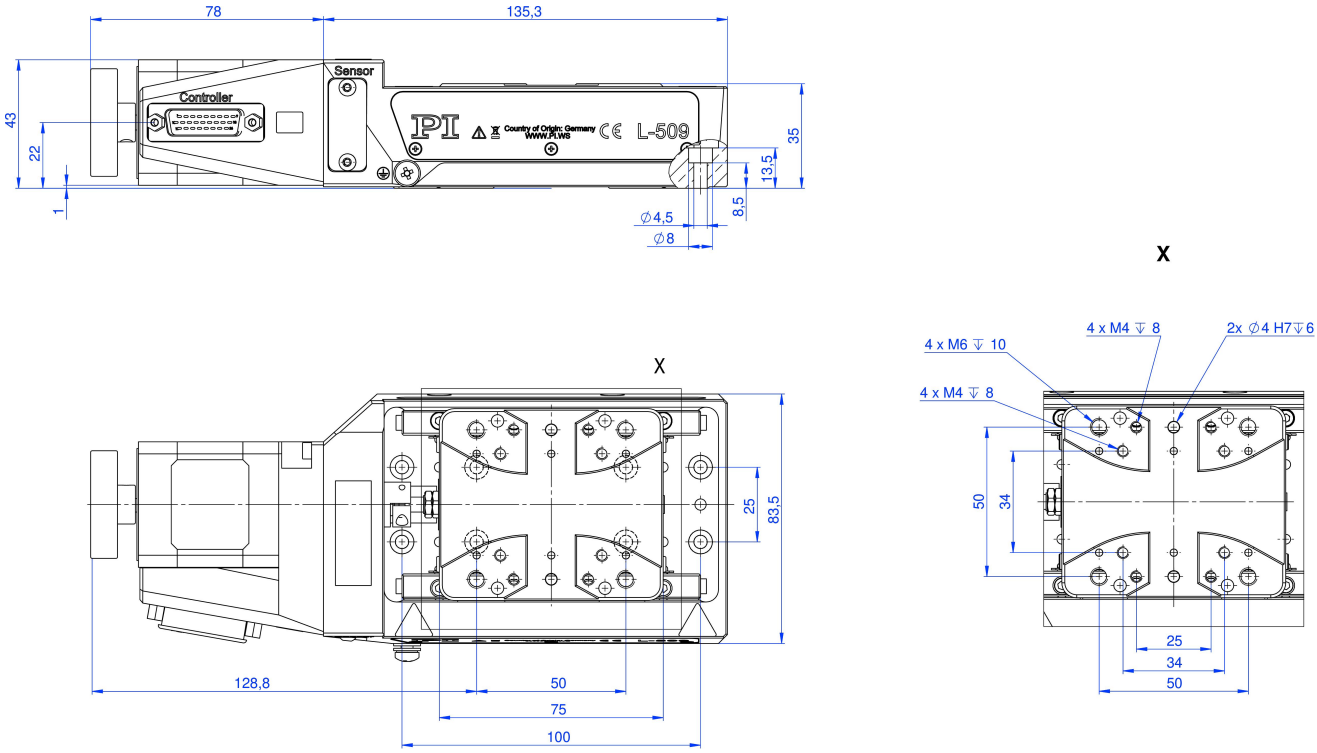
Drive properties	Unit	Tolerance	L-509.051300V7	L-509.031600V9	L-509.051600V9
Drive type			2-phase stepper motor	2-phase stepper motor	2-phase stepper motor
Nominal voltage	V		24	24	24
Maximum power consumption	W		10	10	10
Motor resolution	Full steps/ rev.		200	200	200
Drive force in positive direction of motion in X	N	Typ.	60	60	60
Drive force in negative direction of motion in X	N	Typ.	60	60	60
Resistance phase-phase	Ω	Typ.	1.7	1.7	1.7
Inductance phase-phase	mH		3	3	3
Back EMF, phase-phase, rotational	V/kRPM	Max.	—	—	—

Mechanical properties	Unit	Tolerance	L-509.051300V7	L-509.031600V9	L-509.051600V9
Permissible push force in Y	N	Max.	50	50	50
Permissible push force in Z	N	Max.	100	100	100
Permissible torque in θX	N·m	Max.	30	30	30
Permissible torque in θY	N·m	Max.	25	25	25
Permissible torque in θZ	N·m	Max.	20	20	20
Holding force in X, passive	N		50	30	30
Moved mass in X, unloaded	g		220	220	220
Drive screw type			Ball screw	Ball screw	Ball screw
Drive screw pitch	mm		1	1	1
Gear ratio i			—	—	—
Guide			Crossed roller guide	Crossed roller guide	Crossed roller guide
Overall mass	g		1600	1400	1600
Material			Aluminum, steel	Aluminum, steel	Aluminum, steel

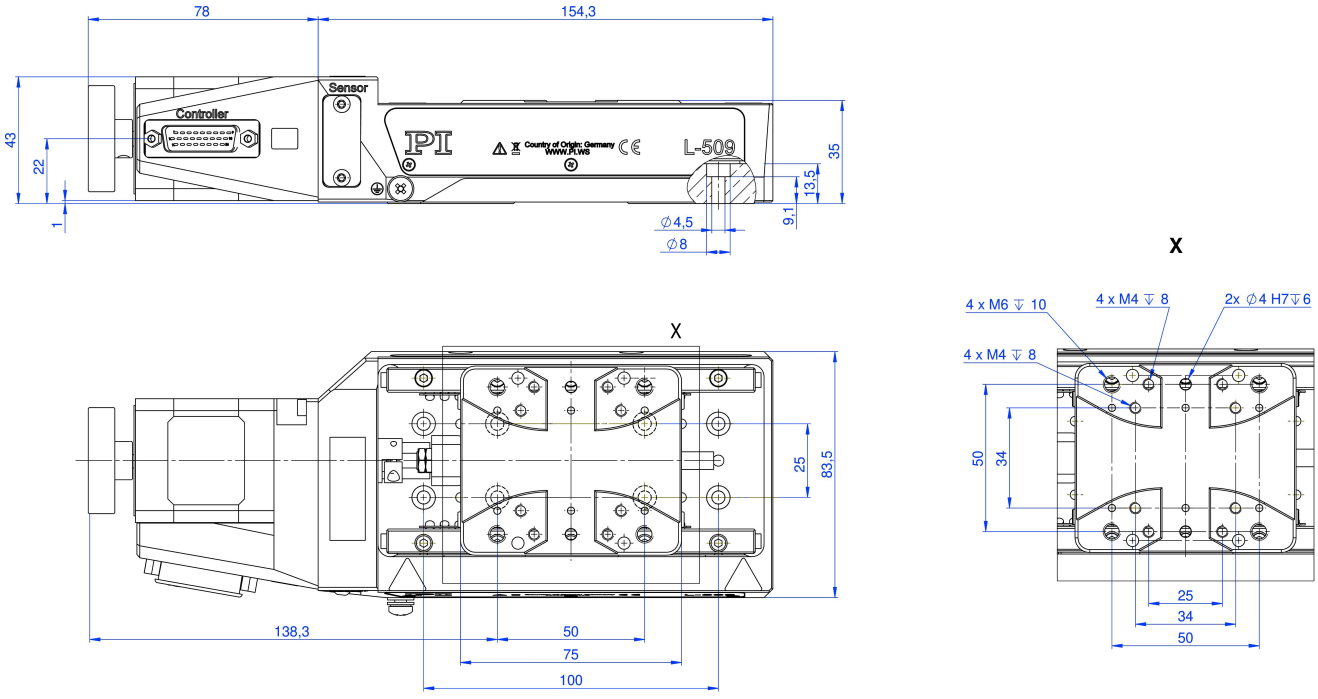
Miscellaneous	Unit		L-509.051300V7	L-509.031600V9	L-509.051600V9
Operating temperature range	°C		5 to 40	5 to 40	5 to 40
Vacuum class	hPa		10^{-7}	10^{-9}	10^{-9}
Maximum bakeout temperature	°C		80	150	150
Connector			D-sub 15 (f)	D-sub 15 (f)	D-sub 15 (f)
Cable length	m		2	2	2
Recommended controllers/ drivers			C-663.12 C-885 with C-663.12C885 G-901 G-910 ACS modular controller	C-663.12 C-885 with C-663.12C885 G-901 G-910 ACS modular controller	C-663.12 C-885 with C-663.12C885 G-901 G-910 ACS modular controller

At PI, technical data is specified at 22 ±3 °C. Unless otherwise stated, the values are for unloaded conditions. Some properties are interdependent. The designation "typ." indicates a statistical average for a property; it does not indicate a guaranteed value for every product supplied. During the final inspection of a product, only selected properties are analyzed, not all. Please note that some product characteristics may deteriorate with increasing operating time.

Drawings / Images

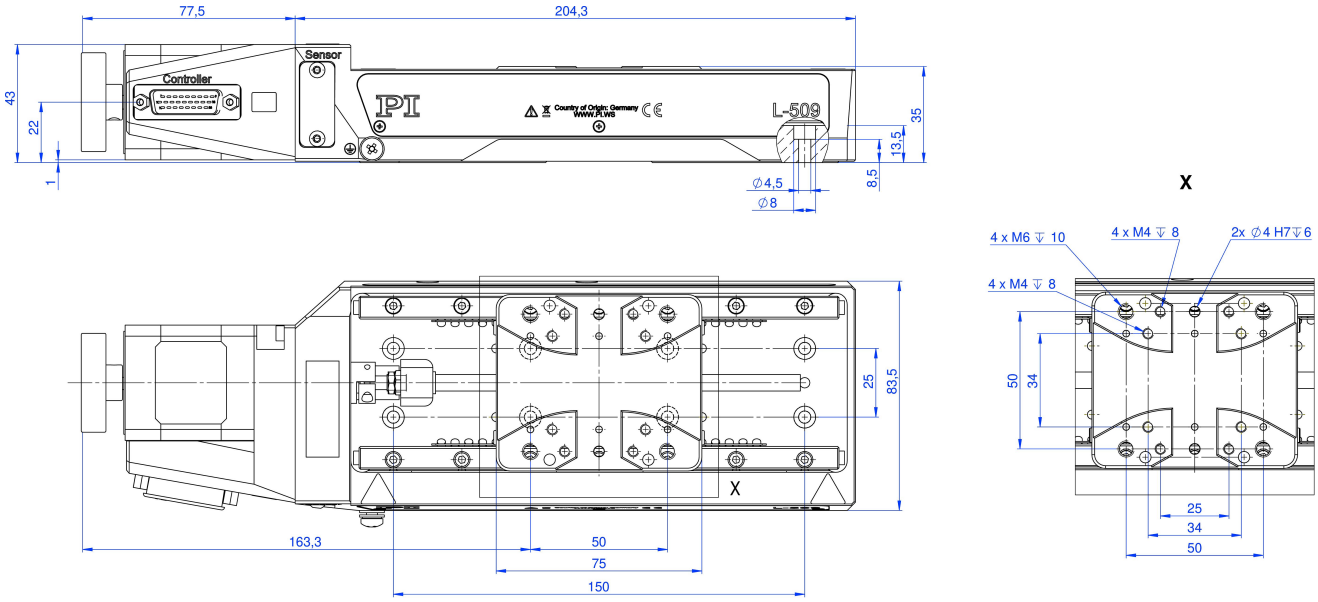


L-509.021100V6, dimensions in mm. Note that a comma is used in the drawings instead of a decimal point.

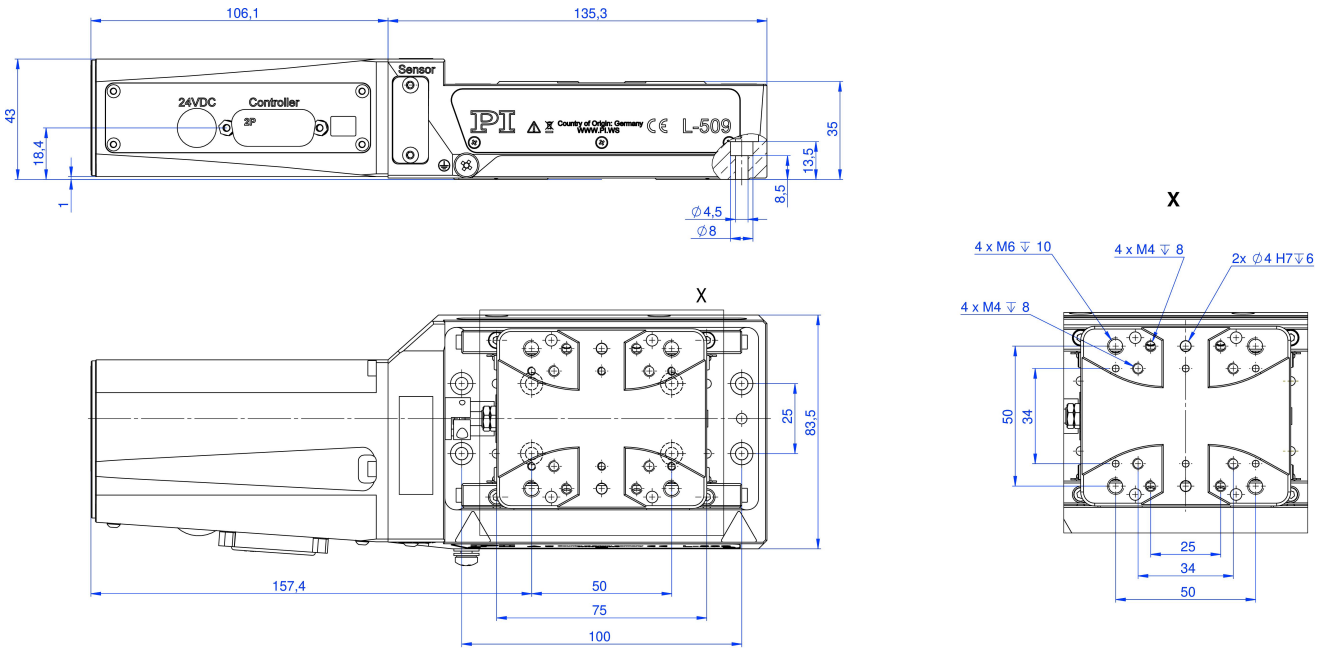


L-509.031100V6, dimensions in mm. Note that a comma is used in the drawings instead of a decimal point.

Drawings / Images

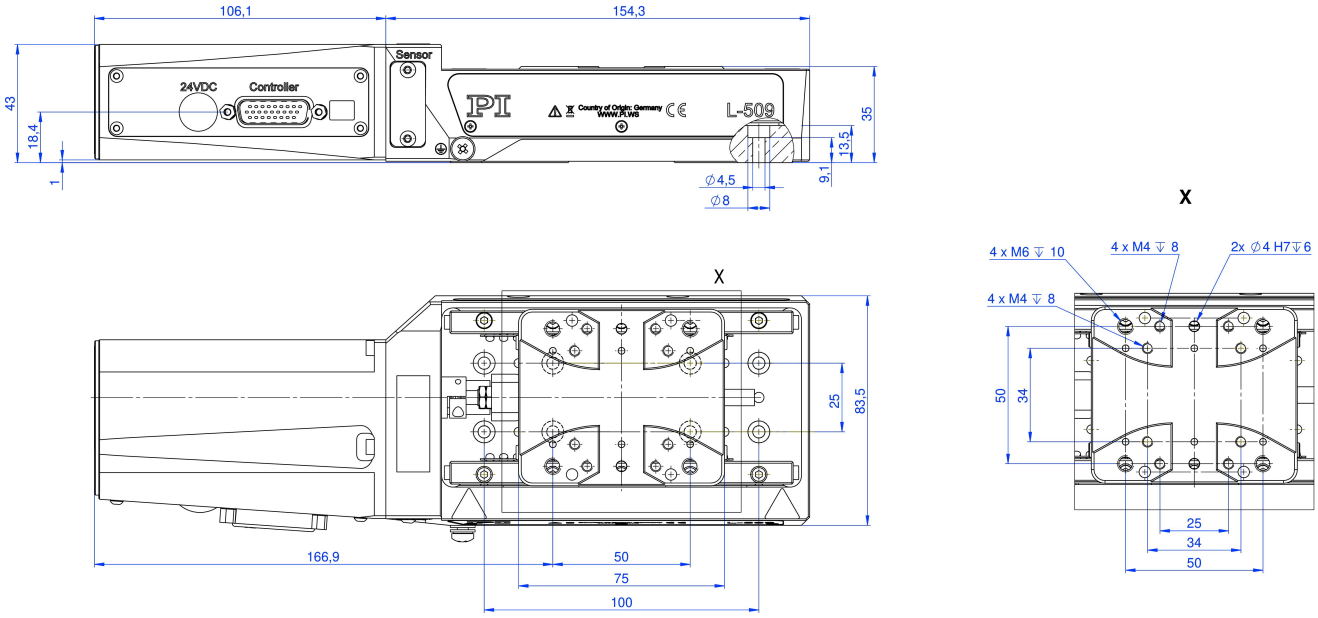


L-509.051100V6, dimensions in mm. Note that a comma is used in the drawings instead of a decimal point.

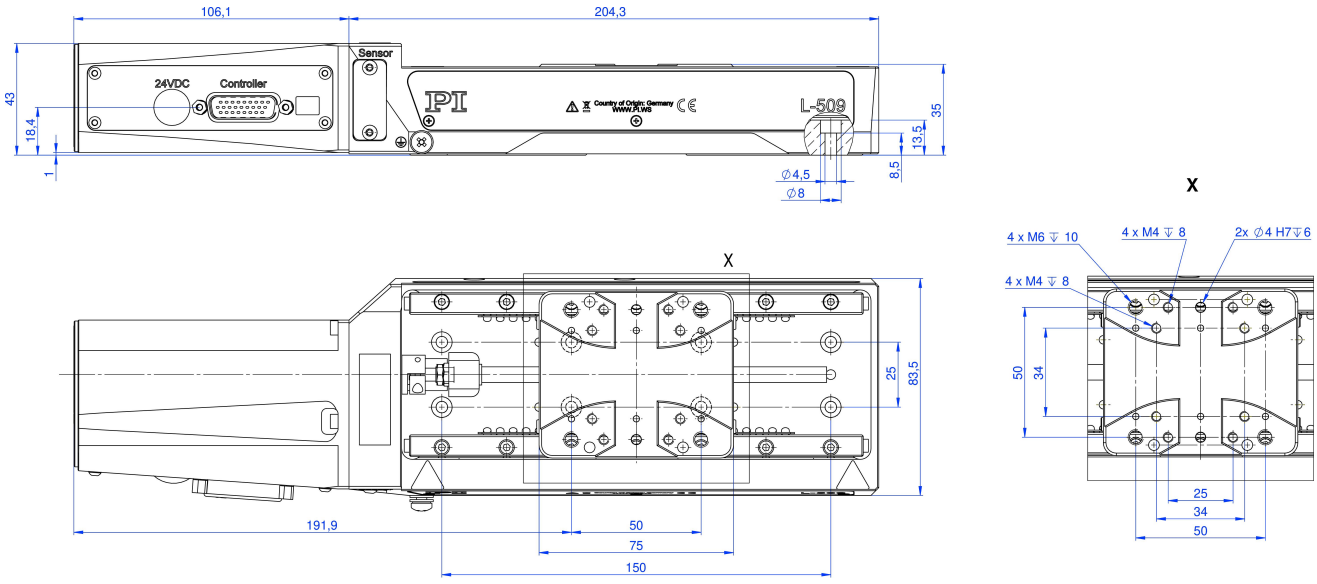


L-509.024132V6, dimensions in mm. Note that a comma is used in the drawings instead of a decimal point.

Drawings / Images

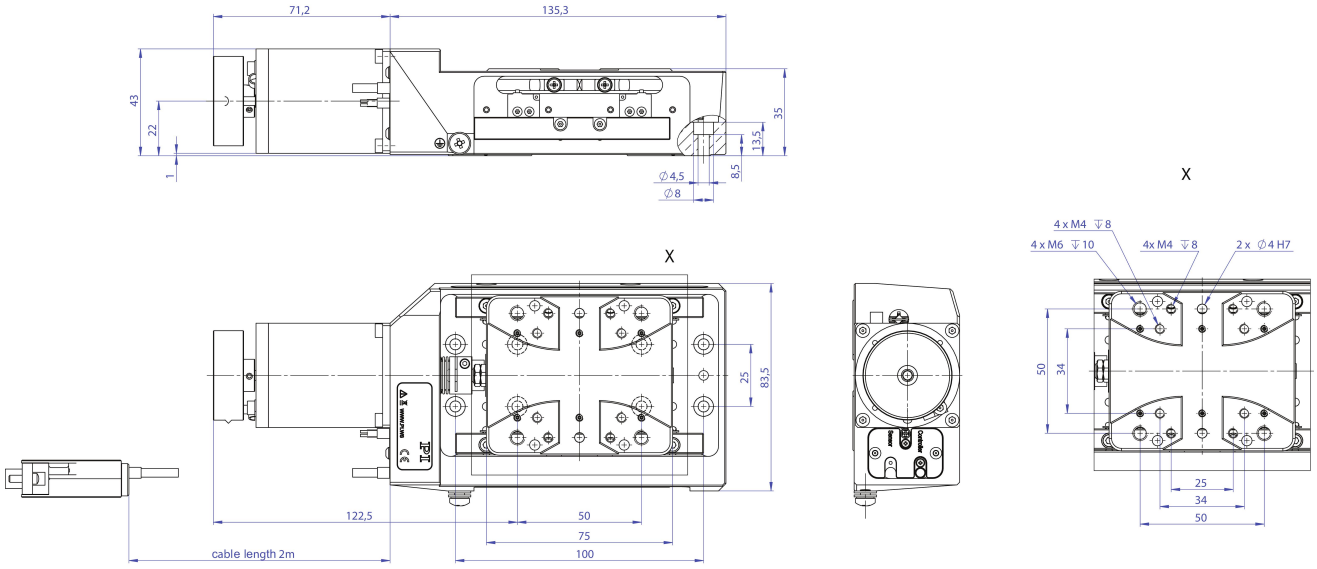


L-509.034132V6, dimensions in mm. Note that a comma is used in the drawings instead of a decimal point.

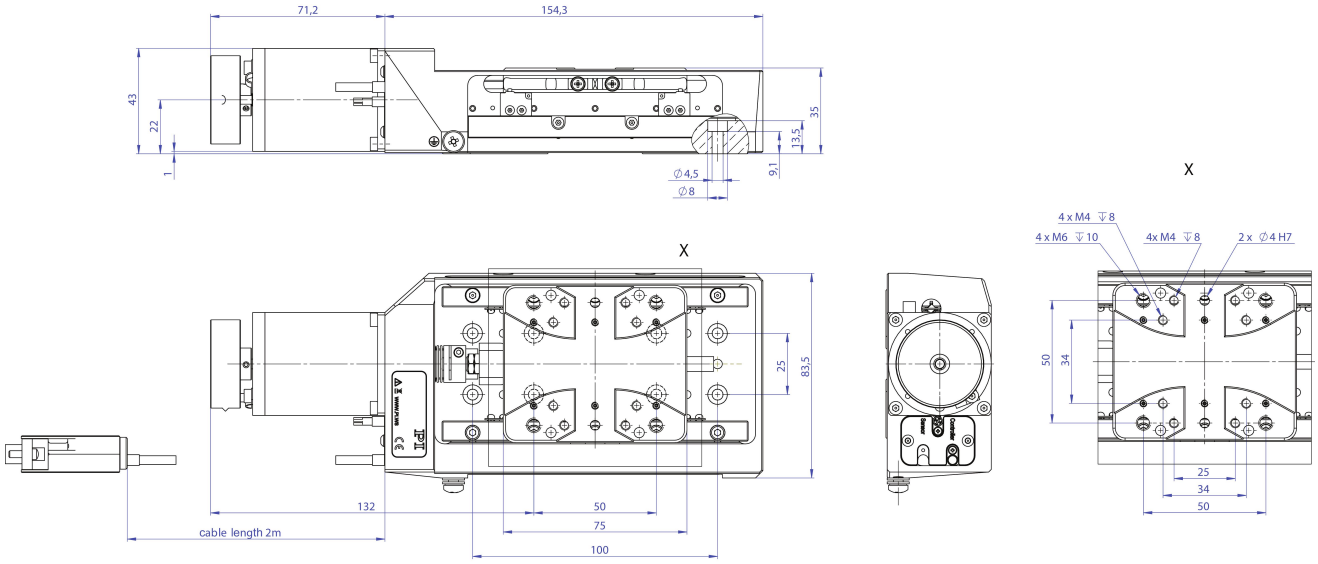


L-509.054132V6, dimensions in mm. Note that a comma is used in the drawings instead of a decimal point.

Drawings / Images

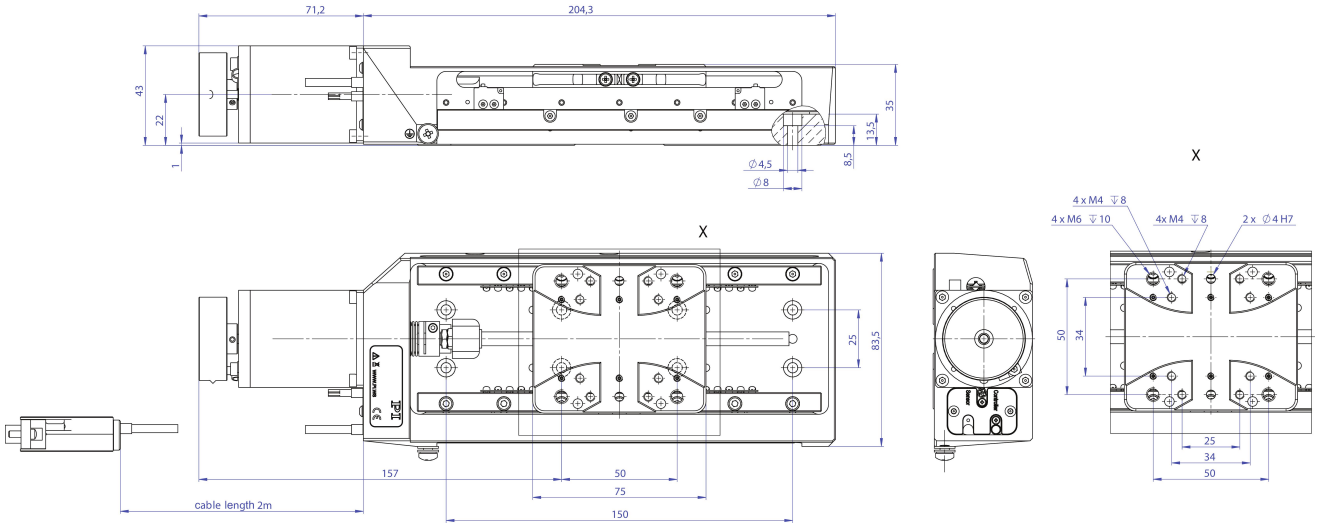


L-509.021300V7, dimensions in mm. Note that a comma is used in the drawings instead of a decimal point.



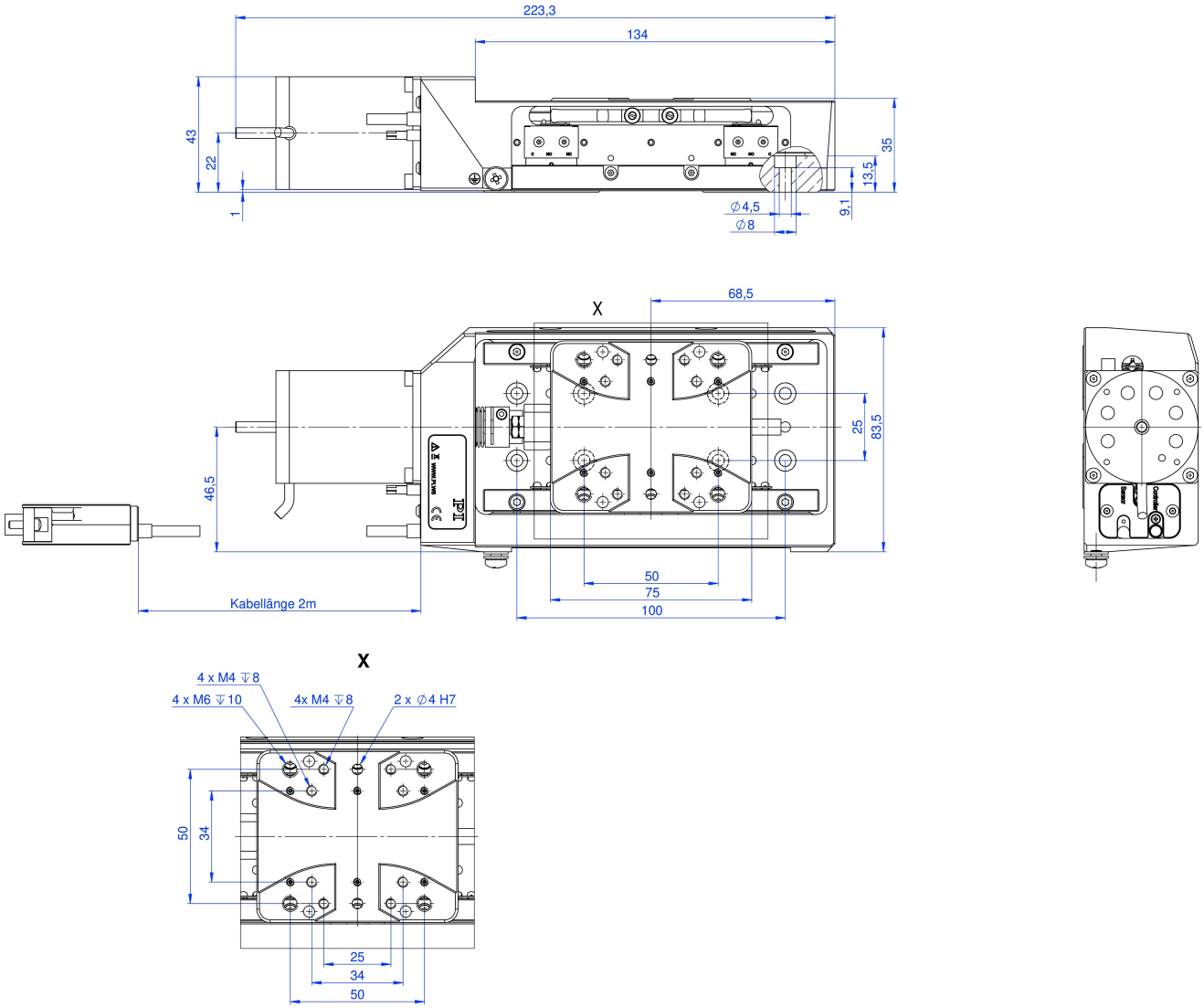
L-509.031300V7, dimensions in mm. Note that a comma is used in the drawings instead of a decimal point.

Drawings / Images



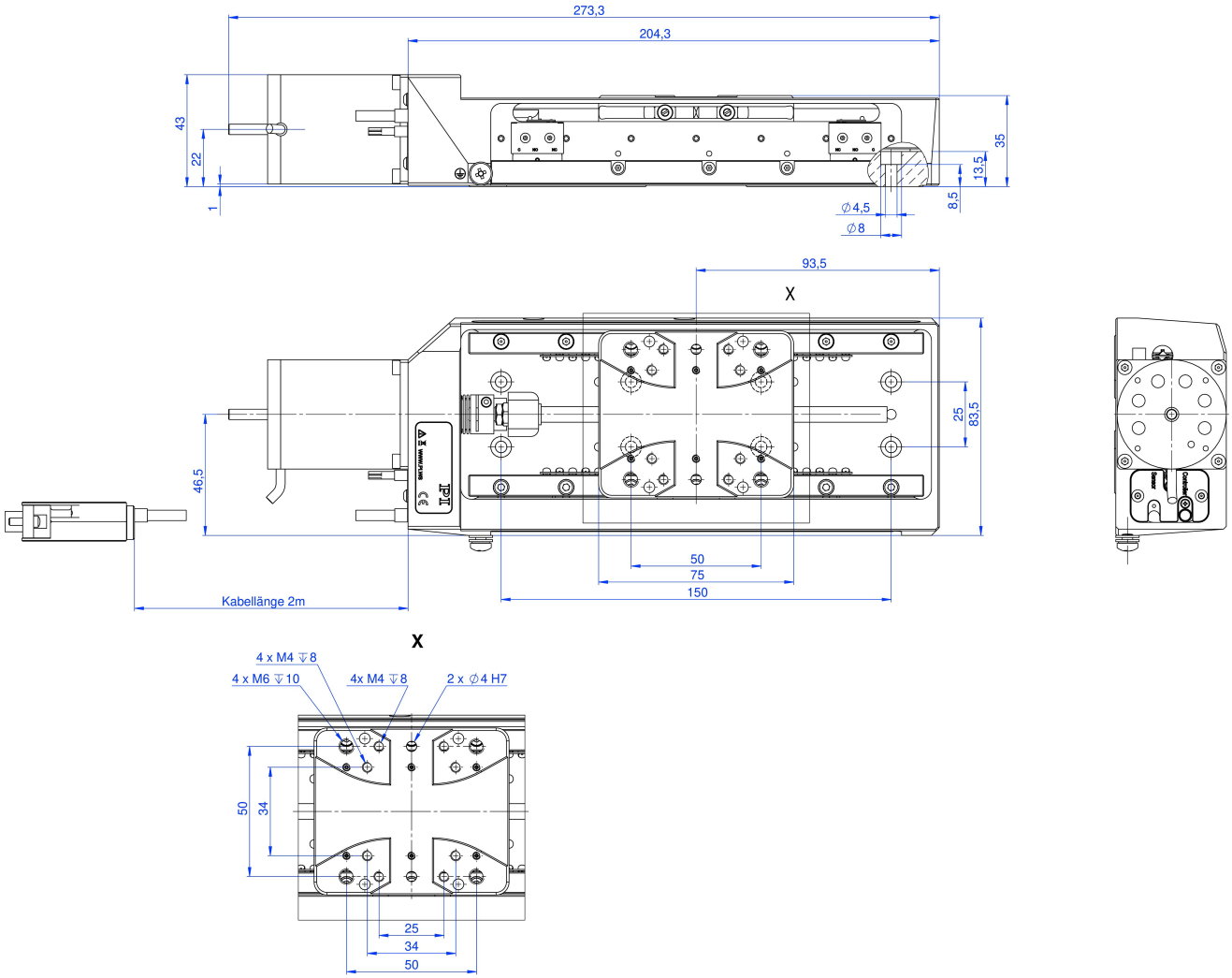
L-509.051300V7, dimensions in mm. Note that a comma is used in the drawings instead of a decimal point.

Drawings / Images



L-509.031600V9, dimensions in mm. Note that a comma is used in the drawings instead of a decimal point.

Drawings / Images



L-509.051600V9, dimensions in mm. Note that a comma is used in the drawings instead of a decimal point.

Order Information

L-509.021100V6
Vacuum-compatible precision linear stage; 2-phase stepper motor; 26 mm travel range; 100 N load capacity; 10 mm/s maximum velocity; ball screw; vacuum compatible to 10 hPa

L-509.031100V6
Vacuum-compatible precision linear stage; 2-phase stepper motor; 52 mm travel range; 100 N load capacity; 10 mm/s maximum velocity; ball screw; vacuum compatible to 10 hPa

Order Information

L-509.051100V6
Vacuum-compatible precision linear stage; 2-phase stepper motor; 102 mm travel range; 100 N load capacity; 10 mm/s maximum velocity; ball screw; vacuum compatible to 10 hPa

L-509.024132V6
Vacuum compatible precision linear stage; DC gear motor; 26 mm travel range; 100 N load capacity; 1.5 mm/s maximum velocity; ball screw; incremental rotary encoder; A/B quadrature, RS-422; vacuum compatible to 10 hPa

L-509.034132V6
Vacuum-compatible precision linear stage; DC gear motor; 52 mm travel range; 100 N load capacity; 1.5 mm/s maximum velocity; ball screw; incremental rotary encoder; A/B quadrature, RS-422; vacuum compatible to 10 hPa

L-509.054132V6
Vacuum compatible precision linear stage; DC gear motor; 102 mm travel range; 100 N load capacity; 1.5 mm/s maximum velocity; ball screw; incremental rotary encoder; A/B quadrature, RS-422; vacuum compatible to 10 hPa

L-509.021300V7
Vacuum-compatible precision linear stage; 2-phase stepper motor; 26 mm travel range; 100 N load capacity; 10 mm/s maximum velocity; ball screw; vacuum compatible to 10 hPa; 2 m cable length

L-509.031300V7
Vacuum-compatible precision linear stage; 2-phase stepper motor; 52 mm travel range; 100 N load capacity; 10 mm/s maximum velocity; ball screw; vacuum compatible to 10 hPa; 2 m cable length

L-509.051300V7
Vacuum-compatible precision linear stage; 2-phase stepper motor; 102 mm travel range; 100 N load capacity; 10 mm/s maximum velocity; ball screw; vacuum compatible to 10 hPa; 2 m cable length

L-509.031600V9
Vacuum-compatible precision linear stage; 2-phase stepper motor; 52 mm travel range; 100 N load capacity; 10 mm/s maximum velocity; ball screw; vacuum compatible to 10 hPa; 2 m cable length

L-509.051600V9
Vacuum-compatible precision linear stage; 2-phase stepper motor; 102 mm travel range; 100 N load capacity; 10 mm/s maximum velocity; ball screw; vacuum compatible to 10 hPa; 2 m cable length